

Louder, Longer, Livelier: Acoustic Shortcuts and Underspecified Rationales in Speech LLM Judges

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Abstract

LLM-as-a-judge is widely used for evaluating text, but extending this paradigm to speech requires models to interpret acoustic as well as linguistic evidence. This introduces a modality-specific risk: a speech judge may treat a perceptually salient cue as evidence of quality even when that cue is irrelevant to the target criterion or receives more weight than human listeners give it. We call this behavior an *acoustic shortcut*. To study it, we audit six speech LLM judges using controlled manipulations of intensity, content richness, and emotional delivery. We evaluate both pointwise scoring and pairwise comparison, using human preference calibration to interpret the results. The judges consistently reward louder audio, prefer content-rich speech more strongly than human listeners do, and map emotional delivery into quality preferences. These effects are most visible in pairwise comparison, while pointwise scores often obscure them. More concerning, the accompanying rationales rarely identify the acoustic cue that changes a judgment and instead repeatedly rely on a limited vocabulary, leaving them acoustically underspecified. Together, these findings show that reliable speech

judges must both resist acoustic shortcuts and ground their rationales in the acoustic evidence behind their decisions. To support reproducibility and future audits, we also release **SPEECH-JUDGEAUDIT**, the controlled stimuli and evaluation tools used in this study.

1 Introduction

Speech quality is multidimensional and requires fine-grained acoustic perception. Listeners may attend to clarity, intelligibility, pleasantness, listening effort, naturalness, and other attributes depending on the task and context (Loizou, 2011). Human listening tests such as Mean Opinion Score (MOS) and pairwise studies remain the most direct way to measure these judgments, but they are slow, expensive, and difficult to run at every stage of speech technology development (Wester et al., 2015; Kirkland et al., 2023). Neural MOS predictors support faster iteration, but they are usually trained to reproduce a particular scalar rating, provide little information about which acoustic evidence shaped that score, and may not generalize to new listening-test contexts (Cooper et al., 2022).

Speech LLM judges offer another way to scale evaluation. They range from specialized large

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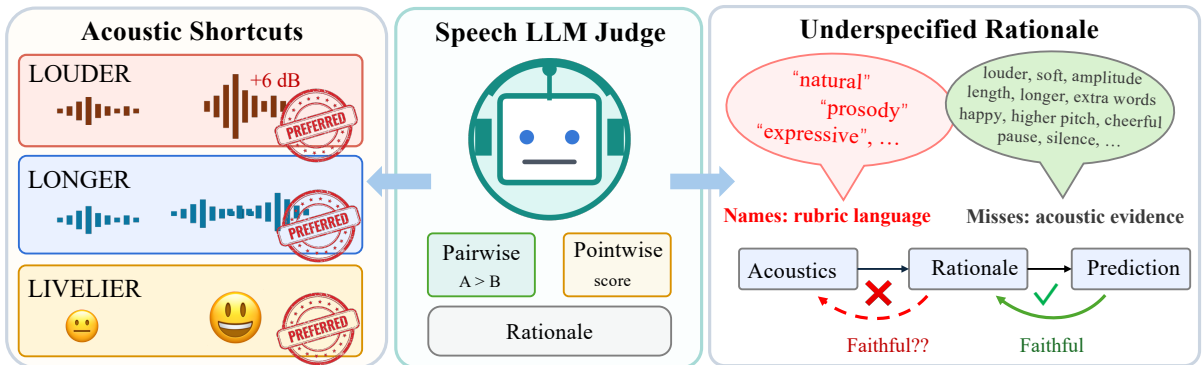


Figure 1: Speech LLM judges may turn salient acoustic cues such as louder, longer, or livelier speech into quality preferences, while their rationales often use broad rubric terms rather than naming the cue that changed the judgment.

audio language models (LALMs) fine-tuned for speech-quality evaluation to proprietary, general-purpose LALMs evaluated with zero-shot prompts. They can listen to one or two audio clips and return a score, preference, or descriptive rationale. Specialized judges also differ in scope: SpeechJudge targets naturalness (Zhang et al., 2025), while SQ-LLM (Wang et al., 2026a) and UniSRM (Wang et al., 2026b) assess multiple dimensions of overall quality, including intelligibility, prosody, and listening effort. Their outputs can support data selection and inference-time evaluation, or serve as reward signals for training speech models (Zhang et al., 2025; Ji et al., 2025). Consequently, biases in these judgments can affect both leaderboard results and models trained with judge feedback.

This raises a fundamental question: do speech LLM judges evaluate the intended quality reliably, or do they let salient acoustic cues become shortcuts? Text-based LLM judges are known to favor candidates based on their position, length, or model family (Zheng et al., 2023; Wang et al., 2023; Saito et al., 2023; Panickssery et al., 2024). Speech LLM judges may inherit these risks, while the audio modality introduces a distinct, modality-specific risk: speech varies in pitch, loudness, pauses, speaking rate, emotional delivery, and more, and a judge must determine which of these cues are relevant to the quality being evaluated.

To study this risk, we examine whether judges use acoustic cues in ways justified by the intended quality criterion. **We call this behavior an *acoustic shortcut* when an easy-to-detect acoustic cue shapes a quality judgment despite being irrelevant to the target criterion or receiving much more weight than human listeners give it.** We audit six speech LLM judges using both human recordings and generated speech. Although one audited model is a LALM-based scalar reward model rather than a generative rationale-producing judge, we include it as a speech-LLM-based evaluator. Our controlled comparisons isolate one factor at a time, allowing us to test whether each manipulation systematically affects final judgments and, when it does, whether the accompanying rationales identify the relevant acoustic change. We use human preference calibration when a factor may reasonably affect perceived quality. Our contributions and findings are:

- **A controlled acoustic audit of speech LLM judges.** We audit six judges using controlled

manipulations of intensity, content richness, and emotional delivery, with human calibration when a factor may affect perceived quality. We test whether controlled acoustic changes affect pointwise and pairwise judgments and whether rationales name the relevant acoustic changes. We release the resulting stimuli and evaluation tools as SPEECH-JUDGEAUDIT, allowing the same tests to be applied to other speech judges. Section 3 describes the methodology.

- **Judges turn salient acoustic cues into quality preferences.** We find that judges reward louder audio even when only signal level changes, prefer content-rich speech more strongly than human listeners do, and map emotional delivery into quality preferences. These effects are clearest in pairwise comparison, while pointwise scores often make them less visible. Section 4 presents the results. Such judges can reward systems for changing easy-to-detect acoustic properties instead of improving the intended quality.
- **Rationales are acoustically underspecified.** We find that rationales repeatedly draw on a limited vocabulary across acoustically different inputs. Even when a manipulation changes the judgment, the rationales rarely name the manipulated acoustic cue, leaving them acoustically underspecified. Detailed results are presented in Section 5. This limits their value as debugging signals for speech evaluation and model development.
- **Implications for using and training speech judges.** The audit also supports practical guidance for speech evaluation: control salient acoustic properties before judging, compare pointwise and pairwise behavior when possible, and train future judges with rationale data grounded in localized acoustic evidence rather than only broad rubric language. Section 6 discusses these implications.

2 Related Work

2.1 Speech LLM Judges

Recent work has explored large audio language models (LALMs) as scalable evaluators of speech. One line of work uses general-purpose LALMs

without task-specific fine-tuning. AudioJudge evaluates zero-shot speech assessment with general-purpose audio models and documents reliability issues such as position and transcript-length bias (Manakul et al., 2026). TRACE instead converts acoustic measurements into a structured written description before asking a text LLM to evaluate the speech (Chandra et al., 2026).

Another line of work develops specialized judges by fine-tuning existing LALMs, such as Qwen2.5-Omni (Xu et al., 2025), for speech-quality evaluation. SpeechJudge focuses on naturalness and uses human preference data to train both a scalar Bradley–Terry model and a pairwise judge that also produces rationales (Bradley and Terry, 1952; Zhang et al., 2025). The Bradley–Terry baseline, SpeechJudge-BTRM, adds a linear reward head to Qwen2.5-Omni-7B and outputs a single scalar reward; unlike SpeechJudge-GRM, it does not generate rationales. SQ-LLM and UniSRM also build on the same backbone but target broader speech-quality and reward-modeling settings (Wang et al., 2026a,b). Related work further broadens speech evaluation to low-level quality descriptions, spoken interaction, and instruction-driven evaluation (Wang et al., 2025; Ji et al., 2025; Shen et al., 2026; Zhang et al., 2026).

These studies show that speech LLM judges can return scores, preferences, and natural-language rationales. However, prior work has not systematically tested whether speech judges treat controlled acoustic cues as invalid or overweighted evidence of quality. Our work addresses this question through controlled acoustic manipulations across both general-purpose and specialized judges.

2.2 From LLM-Judge Bias to Acoustic Shortcuts

Research on text-based LLM judges provides a starting point for understanding this problem. These judges can favor candidates based on position, length, verbosity, or model identity rather than the target quality (Wang et al., 2023; Zheng et al., 2023; Singhal et al., 2023; Saito et al., 2023; Dubois et al., 2024; Panickssery et al., 2024; Saito et al., 2024). When judge outputs are used as training rewards, such biases may lead to reward hacking (Gao et al., 2022). Together, these are instances of shortcut learning, in which an easily observed property shapes a judgment more than the quality being evaluated (Geirhos et al., 2020).

The audio modality broadens the set of possi-

ble shortcuts because speech carries linguistic content alongside loudness, pitch, duration, silence, speaking rate, emotional delivery, and other acoustic properties (Loizou, 2011). Controlled studies show that LALMs can favor easier linguistic or contextual cues over acoustic evidence in emotion, paralinguistic, and speaker-consistency tasks (Chen et al., 2025; Pang et al., 2026; Lee et al., 2026). Although these studies do not directly evaluate speech judges, they motivate auditing how specific acoustic cues shape quality judgments.

2.3 Rationale Faithfulness and Acoustic Evidence

Acoustic shortcuts concern what judges decide; rationales raise a second question: whether those decisions are grounded in the audio. To make judgments transparent, rationales should connect decisions to the relevant acoustic evidence.

A growing body of work trains LALMs to produce multi-step rationales for audio tasks, especially audio question answering (Ma et al., 2025; Xie et al., 2025; Kong et al., 2025; Wijngaard et al., 2025; Xiong et al., 2025; Li et al., 2025; He et al., 2026). These studies show that LALMs can produce structured reasoning and that such reasoning can improve task performance.

For general audio reasoning, Mousavi et al. (2025) intervene on model rationales and find that final answers often depend on them. This tests *rationale faithfulness*: whether the answer follows the stated reasoning, regardless of whether that reasoning accurately describes the audio. SpeechJudge reports similar internal consistency in speech judging, with final preferences usually matching the reasons stated in the rationale (Zhang et al., 2025).

These studies mainly test one link: whether the final prediction is faithful to the stated rationale. Our audit asks about the other link: when a controlled acoustic change affects a speech-judge decision, does the rationale connect that decision to the acoustic evidence in the input? A rationale can be consistent with the final preference while still failing to name the cue that changed the judgment.

3 Methodology

We conducted a controlled audit of six speech LLM judges. In each experiment, we varied a targeted property of the speech while holding potential confounds fixed, then examined (1) whether

Judge	Backbone	Quality criterion	Pairwise	Pointwise	Scale	Rationale
SpeechJudge-BTRM (Zhang et al., 2025)	Qwen2.5-Omni	Naturalness	×	✓	$(-\infty, \infty)$	×
SpeechJudge-GRM (Zhang et al., 2025)	Qwen2.5-Omni	Naturalness	✓	×	1–10	✓
UniSRM (Wang et al., 2026b)	Qwen2.5-Omni	Overall quality	✓	✓	1–5	✓
SQ-LLM (Wang et al., 2026a)	Qwen2.5-Omni	Overall quality	✓	✓	1–5	✓
Gemini-2.5-pro (Gemini Team, Google, 2025)	Gemini	Overall quality	✓	✓	1–10	✓
Gemini-3.1-pro (Gemini Team, 2026)	Gemini	Overall quality	✓	✓	1–10	✓

Table 1: Audited speech LLM judges. The Gemini models are general-purpose LALMs evaluated with zero-shot prompts; the other four judges are specialized models fine-tuned for speech-quality evaluation. SpeechJudge-BTRM is a scalar reward model rather than a generative judge, but we include it as a LALM-based speech evaluator. The rationale column indicates whether the judge returns an explanation with its judgment.

the change systematically affected final judgments and (2) when it did, whether the accompanying rationales named the relevant acoustic change.

3.1 Judges and Audit Scope

We selected six recent speech LLM judges, covering specialized models with publicly available weights and general-purpose models evaluated with zero-shot prompts. Their target criteria were naturalness or overall speech quality, and their supported outputs included pointwise scores, pairwise preferences, and rationales (Table 1).

The emotional-delivery experiment used human recordings. The remaining experiments used synthesized stimuli from four text-to-speech (TTS) systems spanning autoregressive and non-autoregressive architectures and both codec-token and mel-spectrogram outputs. We used this diversity to avoid relying on a single synthesis architecture. System details are provided in Appendix A.1.

3.2 Controlled Acoustic Manipulations

We audit three controlled acoustic factors: intensity adjustment, content richness, and emotional delivery. Complete stimulus construction details are provided in Appendix A.2.

Intensity. We selected 80 sentences from LibriTTS test-clean (Zen et al., 2019), with 20 synthesized by each TTS system. We normalized each utterance to -30 LUFS to form the fixed reference. For each shift $g \in \{-6, -3, +3, +6\}$ dB, we scaled the reference signal by $10^{g/20}$, changing its RMS level while holding all other signal properties fixed.

Content richness. We used 149 concise–richer pairs from ASSET (Alva-Manchego et al., 2020). Each pair expressed the same intended meaning and was synthesized with the same TTS system,

speaker, and reference voice. Appendix A.2 gives a representative concise–rich pair.

Duration controls. We used two controls to separate content richness from duration. For each ASSET pair, silence padding matched the richer clip’s duration without adding speech, while repetition duplicated the concise speech without adding new content. Table 2 reports the resulting comparison.

Emotional delivery. We used 800 recordings from ESD (Zhou et al., 2022) where 10 speakers each recorded the same 20 sentences with Neutral, Happy, Sad, and Angry delivery. Within each matched set, speaker and semantic content were held constant, with intended emotional delivery as the experimental factor. Before evaluation, we normalized all recordings to -23 LUFS and trimmed leading and trailing silence.

3.3 Evaluation and Analysis

Judgment analysis. For pointwise evaluation, each clip was scored independently. Because judges used different scales, we compared scores only within the same judge. Pairwise evaluation selected one clip or returned a tie. We tested both A/B and B/A orders to control position effects documented in text and audio judges (Wang et al., 2023; Manakul et al., 2026). We encoded a manipulated-clip win as 1, a reference-clip win as 0, and a tie as 0.5, averaging the two orders within each item before aggregation. Confidence intervals used item-level bootstrap samples. For pointwise emotional-delivery scores, we fit within-judge mixed-effects models with emotion as a fixed effect and random intercepts for speaker and sentence. Prompts, statistical tests, and per-judge results are provided in Appendices A.3 and B.1–B.5.

Human preference calibration. Fifteen native or near-native English listeners were assigned 30 pairs each, yielding 449 valid judgments over 60

unique pairs: 12 intensity, 18 content-richness, 15 Sad–Neutral, and 15 Happy–Neutral. Presentation order was randomized, and listeners selected the first clip, the second clip, or a tie. Their responses provide a calibration baseline: for content richness and emotional delivery, they help distinguish judge-specific overweighting from preferences shared with listeners; for intensity adjustment, they confirm that loudness is perceptually salient, even though it is not valid evidence of better speech quality in our controlled comparison. Full instructions and response statistics are provided in Appendix D.1.

Rationale analysis. The five rationale-producing judges returned either free-form explanations or structured reasoning fields. Rationale analysis asks whether these explanations make changed judgments transparent. We analyze two properties: whether rationales adapt their vocabulary when acoustic inputs change, and whether they explicitly name the manipulated acoustic cue when that cue affects the judgment. Using prespecified lexicons for loudness, silence, repetition, content richness, and emotion, we measure how often rationales name each manipulation. We also compare frequent words across conditions to identify recurring vocabulary. Judge–condition coverage and lexical summaries are provided in Appendix C.1. This analysis concerns what the rationales explicitly say, not the model’s unobserved internal reasoning.

4 Acoustic Shortcuts in Judge Decisions

For each acoustic manipulation, we ask whether it changes speech LLM judge decisions, then compare judge behavior with human preference calibration when the cue may plausibly affect perceived quality. This lets us distinguish cues that are invalid evidence for the target quality from cues that judges overweight relative to human listeners.

4.1 Judges Reward Louder Audio

The intensity manipulation changes only the RMS level of a fixed reference signal (Section 3.2). Pairwise comparison reveals a consistent preference for louder audio. At +6 dB, all five pairwise judges favor the louder clip, with mean preference 0.578 and individual rates from 0.541 to 0.616 (Figure 2).

Pointwise scoring gives a less consistent picture. UniSRM assigns a small but significant increase at +6 dB, whereas SQ-LLM returns identical pointwise scores despite preferring the louder

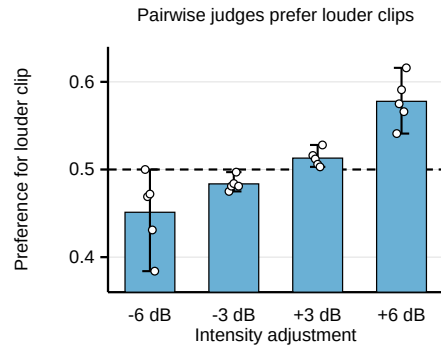


Figure 2: Mean pairwise preference for the louder clip after intensity adjustment. Points show individual judges; full per-judge results are in Appendix B.2.

clip pairwise. The Gemini pointwise differences are not significant, and BTRM penalizes quieter audio without rewarding the +6 dB condition. Thus the directional loudness preference is much clearer in pairwise comparison.

Human listeners also favor the +6 dB clip (0.612; Appendix D.1), confirming that the change is perceptually salient. This does not make loudness valid evidence of better speech quality: the intensity adjustment changes neither the linguistic content nor the underlying speech. The judges can therefore reward louder audio even when quality has not improved. If such a judge is used for model selection or as a reward signal, it could favor systems that increase output loudness without improving the intended quality.

TAKEAWAY. All five pairwise judges favor the +6 dB speech: loudness is a clear acoustic shortcut.

4.2 Judges Overweight Content Richness

In the ASSET content-richness experiment, all five pairwise judges prefer the meaning-equivalent richer clip. Because this manipulation changes more than one surface property, Table 2 separates content richness from duration and repetition controls. The controls do not reproduce the preference: padding or repeating the concise clip does not make it win, while richer speech remains preferred at matched duration. The effect is therefore best interpreted as overweighting richer speech content, rather than a simple preference for longer audio.

Pointwise scores show no consistent richer-speech advantage, and their item-level relationship with pairwise preferences is weak (Appendix B.5). Human listeners are also much less decisive: their mean richer-speech preference is 0.547 and its con-

Comparison	What changes	Mean preference
Silence padding vs. concise	Duration only	0.41
Repetition vs. concise	Duplicated speech	0.46
Richer vs. concise	Content and varied speech material	0.71
Richer vs. duration-matched padding	Content richness at matched duration	0.70

Table 2: Content-richness controls averaged over the five pairwise judges. Values are preferences for the first condition, with ties counted as 0.5. Full per-judge results are in Appendix B.3.

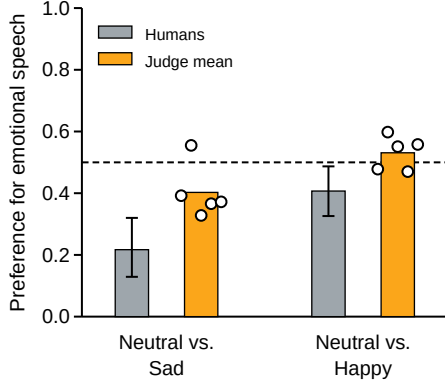


Figure 3: Human and mean judge preference for the emotional clip over the matched Neutral clip. White points show individual pairwise judges.

fidence interval crosses chance, compared with 0.707 for the judges (Appendix D.1). The result is therefore not merely a shared preference for longer utterances. Pairwise judges give content richness substantially more weight than listeners do.

TAKEAWAY. Pairwise judges consistently favor content-rich speech, and the preference is much stronger than in human listeners.

4.3 Emotional Delivery Shapes Quality Judgments

The ESD stimuli match speaker and semantic content across emotional conditions. In pointwise scoring, four of five judges show a significant overall emotion effect. The direction varies by judge: Sad and Angry are often scored lower than Neutral, while Gemini-3.1-pro assigns higher scores to Happy and Angry. In pairwise comparison, Sad loses to Neutral for four of five judges, whereas Happy beats Neutral for UniSRM, SQ-LLM, and Gemini-3.1 (Appendix B.4). Taken alone, the judge audit suggests that changes in intonation, speaking rate, and expressiveness can shape quality preferences in both pointwise and pairwise settings.

Human preference calibration changes how these effects should be interpreted. Listeners strongly prefer Neutral over Sad: the Sad preference rate

is 0.217, compared with a judge mean of 0.403. The Sad penalty is therefore largely shared with listeners rather than a judge-only shortcut. Happy shows a clearer mismatch. The listener mean favors Neutral (Happy preference 0.407), while the judge mean is 0.531 and three judges significantly favor Happy (Figure 3). Angry shows a weaker pattern: GRM and Gemini-2.5 prefer Neutral pairwise, while the other pairwise judges are near chance. Emotional delivery can enter quality judgments, but the judge-specific shortcut is clearer for Happy than for Sad.

TAKEAWAY. Human listeners and judges both penalize Sad speech, while Happy delivery shows the clearer judge-specific shortcut.

4.4 Pairwise Comparison Makes Acoustic Shortcuts More Visible

Across the three main experiments, pointwise scoring and pairwise comparison are not interchangeable. Pairwise comparison consistently exposes the louder and content-richness preferences, while pointwise scores are often weak or resolution-limited. Emotional delivery affects both modes, but pairwise comparison produces more consistent directional preferences across judges (Appendix B.5).

Boundary silence provides an additional diagnostic. UniSRM’s pointwise scores are nearly unchanged when silence is added, but its pairwise preference for the padded clip falls to 0.069 when 1,000 ms of silence is added to each boundary. Other judges show mixed effects or many ties (Appendix B.2), so this is not a universal silence-padding shortcut. It nevertheless illustrates how pairwise comparison can lead a judge to act on an acoustic difference even when its pointwise score barely moves. Pairwise comparison is useful because it exposes small acoustic differences, but the same sensitivity can make an invalid or over-weighted cue more decisive in the final judgment.

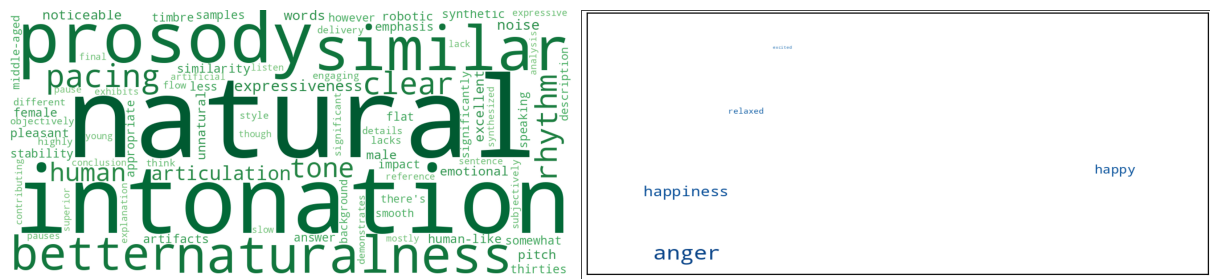


Figure 4: Rationale vocabulary for Neutral and emotional speech. Across acoustically different inputs, rationales repeatedly use similar quality and prosody terms rather than adapting to the specific emotional delivery.

TAKEAWAY. Pairwise comparison exposes acoustic differences, but the same sensitivity can make invalid or overweighted cues more decisive.

5 Rationales Are Acoustically Underspecified

Section 4 shows that controlled acoustic changes can systematically affect judge decisions. We next ask whether the accompanying rationales name the acoustic evidence behind those changes. We separate two problems. Section 5.1 asks whether rationales adapt to different acoustic inputs or fall back on the same quality vocabulary. Section 5.2 asks a narrower question: when an acoustic change affects the judgment, does the rationale say what changed?

5.1 Rationales Draw on a Limited Recurring Vocabulary

Across manipulations, judge rationales repeatedly draw on a limited set of descriptions. Words referring to naturalness, prosody, intonation, clarity, and robotic speech remain common across conditions, while words specific to the manipulation are much rarer. Automatic word-frequency analysis shows this pattern for content richness, intensity, boundary silence, and emotional delivery. Figure 4 illustrates it for Neutral and emotional speech.

This recurring vocabulary can describe the apparent quality of a clip without fully contextualizing the acoustic evidence behind the judgment. For example, when GRM strongly prefers Neutral over Sad delivery for the same speaker and sentence, its rationale is one of the few to identify audible differences: the Sad clip is described as having flatter prosody, slower pacing, and more pauses, while also being called robotic. The rationale is therefore not devoid of acoustic evidence. However, it treats these characteristics directly as naturalness defects without first considering whether they ex-

press a different intended emotion and then assessing naturalness within that delivery. Appendix C.1 provides the full example. This illustrates a subtler grounding gap: a rationale may notice acoustic differences but fail to contextualize them before converting them into quality penalties.

TAKEAWAY. Rationales repeatedly draw on broad quality vocabulary even when the acoustic input changes.

5.2 Rationales Rarely Name the Relevant Acoustic Change

Figure 5 shows the gap between changed judgments and their explanations. Intensity changes affect judgments, yet direct loudness-related terms such as “loud,” “quiet,” and “higher volume” are absent from the audited UniSRM rationales. The same pattern appears for boundary silence and content richness: the relevant words occur in at most a few percent of rationales even when judge preferences change systematically. For emotional delivery, judges rarely name the emotion that was changed; in most cases, the rationale uses generic prosody or quality terms instead of saying whether the clip sounds Happy, Sad, Angry, or Neutral.

As a duration control introduced in Section 3.2, repetition is a partial exception. Four judges mention repetition more often than they name the other manipulations, with rates ranging from 5.7% to 27.9%. Even here, naming is incomplete, and mentioning repetition does not consistently determine whether the repeated clip is penalized (Appendix C.1). The visible rationales can therefore sometimes name a conspicuous acoustic change, but do not do so reliably.

These results do not show that every rationale is factually incorrect or unfaithful to the final answer. They show that visible explanations rely on a limited recurring vocabulary and rarely name the acoustic evidence associated with a systematic change in judgment. This limits the practical

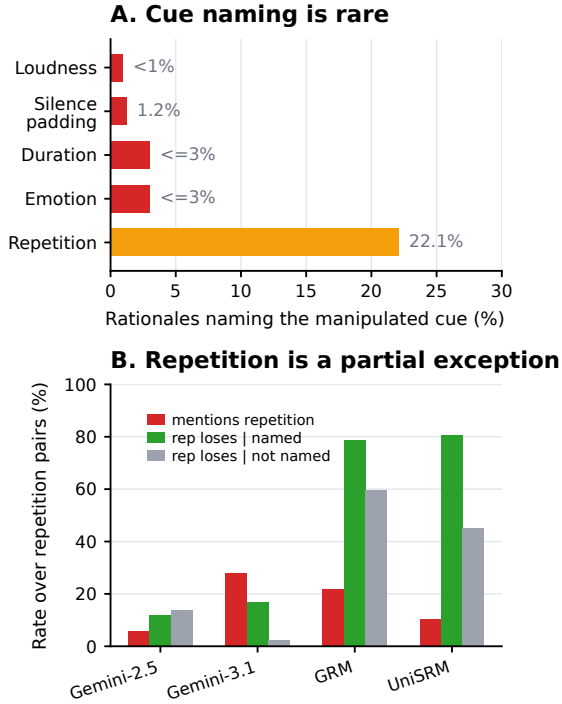


Figure 5: Rationale audit summary. Rationales rarely name the manipulated acoustic cue, while generic rubric terms remain common. Repetition is a partial exception: judges can sometimes name a conspicuous acoustic change, but do not do so reliably.

value of rationales as debugging signals. If a judge changes its decision with loudness, duration, or emotional delivery but explains the decision using generic terms such as naturalness or prosody, the rationale cannot tell users whether the judgment reflects the intended quality or an acoustic shortcut.

TAKEAWAY. When an acoustic change affects a judgment, the rationale usually does not say what changed.

6 Implications

Controlled audits are needed to locate the failure mode. Our supporting analyses show what did not reliably change judgments. Within our stimulus pools, we do not find stable preferences by speaker gender, speaker accent, or Qwen-family self-preference, and broad low-level descriptors do not explain the main effects (Appendix E.1). This does not provide fairness guarantees, but it helps bound the claim: not every salient attribute becomes a shortcut. Rather, specific acoustic cues become shortcuts when the evaluation protocol makes them salient or lets judges give them too much weight.

Using speech judges requires acoustic preprocessing and mode checks. Before using speech LLM judges for system comparison, data selection, or reward modeling, evaluators should normalize loudness and trim leading or trailing silence unless those cues are part of the target quality criterion. Pairwise comparison should also be order-counterbalanced and reported alongside pointwise scoring when both modes are available. Pairwise comparison is useful because it exposes preferences that pointwise scores can hide, but the same contrast can make invalid or overweighted cues more decisive. Human preference calibration is especially important when a cue, such as content richness or emotional delivery, may legitimately affect perceived quality.

Training better judges requires grounded rationale data. The rationale results suggest that simply asking for an explanation is not enough. If supervised fine-tuning data mostly teaches rubric language such as “naturalness,” “prosody,” or “expressiveness,” a judge can produce fluent rationales without identifying the acoustic evidence behind a changed decision. Future speech-judge data should therefore include localized acoustic evidence: the word, phrase, pronunciation, pause, loudness change, emotional delivery, or other audible cue that justifies the judgment. It should also cover diverse speakers, recording conditions, emotions, and synthesis systems so that judges cannot rely on a narrow set of acoustic shortcuts.

Reasoning-oriented training is promising, but should be audited. Recent work shows that reinforcement learning can elicit or improve reasoning behavior in language and speech models (DeepSeek-AI, 2025; Zhang et al., 2025). For speech judges, this suggests a useful direction: training objectives could reward rationales that connect scores to localized acoustic evidence, not only rationales that are fluent or internally consistent with the final preference. However, our findings imply that such training should itself be audited. A rationale can be consistent with the final judgment while still failing to name the acoustic cue that changed the judgment.

Conclusion

We audited six speech LLM judges with controlled acoustic manipulations and human preference calibration. The results show that judges can turn

perceptually salient but invalid or overweighted cues into evidence of quality. They reward louder audio even when only the signal level changes, prefer content-rich speech more strongly than listeners do, and map emotional delivery into quality preferences. These effects are clearest in pairwise comparison, which is useful for exposing small acoustic differences but can also make acoustic shortcuts more decisive.

More concerningly, the accompanying rationales do not reliably make these decisions transparent. Even when an acoustic manipulation changes a judgment, rationales rarely name that change; instead, they often rely on broad rubric terms such as “naturalness” and “prosody,” leaving them acoustically underspecified. Reliable speech judges therefore need two properties: their judgments should be robust to acoustic shortcuts, and their rationales should connect decisions to the acoustic evidence behind them.

7 Limitations

This audit is controlled by design. All stimuli are English; the TTS pools cover four systems and VCTK reference speakers; and the emotion pool uses ESD recordings. These choices let us isolate specific cues, but they do not cover all languages, accents, recording conditions, speakers, or modern TTS systems. Null results for gender, accent, or low-level acoustic descriptors should therefore be read as within-pool nulls, not broad fairness guarantees.

The judges also target different quality criteria. BTRM and GRM focus on naturalness, whereas UniSRM, SQ-LLM, and Gemini prompts target broader overall quality. We define bias relative to the intended quality criterion, but content richness and emotional delivery may reasonably affect perceived quality. This is why we calibrate them against human preference judgments rather than treating every shift as bias by definition.

The human preference study is a calibration, not a full subjective benchmark. It covers only the headline conditions and is large enough to separate the main judge–listener mismatches, but not to estimate fine-grained human preferences across many languages, speakers, emotions, and synthesis domains.

Finally, the rationale audit is lexical. A judge may internally represent a cue without naming it, and rubric words such as “prosody” can sometimes

refer to multiple acoustic properties. Our claim is therefore limited to user-visible rationales: they are acoustically underspecified because they do not reliably and explicitly identify the manipulated acoustic change. We do not infer from this omission that the cue is absent from the judge’s internal representation.

References

- Fernando Alva-Manchego, Louis Martin, Antoine Bordes, Carolina Scarton, Benoît Sagot, and Lucia Specia. 2020. [ASSET: A dataset for tuning and evaluation of sentence simplification models with multiple rewriting transformations](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4668–4679.
- Ralph Allan Bradley and Milton E. Terry. 1952. [Rank analysis of incomplete block designs: I. the method of paired comparisons](#). *Biometrika*, 39(3/4):324–345.
- Arjun Chandra, Kevin J. Miller, Venkatesh Ravichandran, Constantinos Papayiannis, and Venkatesh Saligrama. 2026. [Dimension-first evaluation of speech-to-speech models with structured acoustic cues](#). *Preprint*, arXiv:2601.13742.
- Jingyi Chen, Zhimeng Guo, Jiyun Chun, Pichao Wang, Andrew Perrault, and Micha Elsner. 2025. [Do audio LLMs really LISTEN, or just transcribe? measuring lexical vs. acoustic emotion cues reliance](#). *Preprint*, arXiv:2510.10444.
- Erica Cooper, Wen-Chin Huang, Tomoki Toda, and Junichi Yamagishi. 2022. [Generalization ability of mos prediction networks](#). In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8442–8446. IEEE.
- DeepSeek-AI. 2025. [DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning](#). *Preprint*, arXiv:2501.12948.
- Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. [Length-controlled AlpacaEval: A simple way to debias automatic evaluators](#). *Preprint*, arXiv:2404.04475.
- Leo Gao, John Schulman, and Jacob Hilton. 2022. [Scaling laws for reward model overoptimization](#). *Preprint*, arXiv:2210.10760.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. [Shortcut learning in deep neural networks](#). *Nature Machine Intelligence*, 2:665–673.
- Gemini Team. 2026. [Gemini 3.1 pro model card](#). Published February 2026.

- Gemini Team, Google. 2025. [Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities](#). *Preprint*, arXiv:2507.06261.
- Xiang He, Chenxing Li, Jinting Wang, Yan Rong, Tianxin Xie, Wenfu Wang, Li Liu, and Dong Yu. 2026. [Audio-deepthinker: Progressive reasoning-aware reinforcement learning for high-quality chain-of-thought emergence in audio language models](#). *Preprint*, arXiv:2604.18187.
- Shengpeng Ji, T. Jake Liang, Yangzhuo Li, Jialong Zuo, Minghui Fang, Jinzheng He, Yifu Chen, Zhengqing Liu, Ziyue Jiang, Xize Cheng, et al. 2025. [Wavreward: Spoken dialogue models with generalist reward evaluators](#). *Preprint*, arXiv:2505.09558.
- Ambika Kirkland, Shivam Mehta, Harm Lameris, Gustav Eje Henter, Eva Szekely, and Joakim Gustafson. 2023. [Stuck in the mos pit: A critical analysis of mos test methodology in tts evaluation](#). In *12th Speech Synthesis Workshop (SSW) 2023*.
- Zhifeng Kong, Arushi Goel, João Felipe Santos, Sreyan Ghosh, Rafael Valle, Ping Wei, and Bryan Catanzaro. 2025. [Audio flamingo sound-CoT technical report: Improving chain-of-thought reasoning in sound understanding](#). *Preprint*, arXiv:2508.11818.
- Jonggeun Lee, Junseong Pyo, Gyuhyeon Seo, and Yohan Jo. 2026. [Speakersleuth: Can large audio-language models judge speaker consistency across multi-turn dialogues?](#) *Preprint*, arXiv:2601.04029. Accepted at ACL 2026 Main.
- Gang Li, Jizhong Liu, Heinrich Dinkel, Y. Niu, Junbo Zhang, and Jian Luan. 2025. [Reinforcement learning outperforms supervised fine-tuning: A case study on audio question answering](#). *Preprint*, arXiv:2503.11197.
- Philipos C. Loizou. 2011. [Speech quality assessment](#). In Weisi Lin et al., editors, *Multimedia Analysis, Processing and Communications*, pages 623–654. Springer.
- Ziyang Ma, Zhuo Chen, Yuping Wang, Eng Siong Chng, and Xie Chen. 2025. [Audio-CoT: Exploring chain-of-thought reasoning in large audio language model](#). *Preprint*, arXiv:2501.07246.
- Potsawee Manakul, Woody Haosheng Gan, Michael J. Ryan, Ali Sartaz Khan, Warit Sirichotedumrong, Kunat Pipatanakul, William Barr Held, and Diyi Yang. 2026. [Audiojudge: Understanding what works in large audio model based speech evaluation](#). In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3644–3663, Rabat, Morocco. Association for Computational Linguistics.
- Pooneh Mousavi, Lovenya Jain, Mirco Ravanelli, and Cem Subakan. 2025. [Investigating faithfulness in large audio language models](#). *Preprint*, arXiv:2509.22363.
- Jiacheng Pang, Ashutosh Chaubey, and Mohammad Soleymani. 2026. [Do audio LLMs listen or read? analyzing and mitigating paralinguistic failures with VoxParadox](#). *Preprint*, arXiv:2605.27772.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. [LLM evaluators recognize and favor their own generations](#). In *Advances in Neural Information Processing Systems*.
- K. Saito, Akifumi Wachi, Koki Wataoka, and Youhei Akimoto. 2023. [Verbosity bias in preference labeling by large language models](#). *Preprint*, arXiv:2310.10076.
- K. Saito, Koki Wataoka, and Youhei Akimoto. 2024. [Self-preference bias in LLM-as-a-judge](#). *Preprint*, arXiv:2410.21819.
- Maohao Shen, Tejas Jayashankar, Osama Hanna, Naoyuki Kanda, Yancheng Wang, Kateřina Žmolíková, Ruiming Xie, Niko Moritz, Anfeng Xu, Yashesh Gaur, et al. 2026. [GSRM: Generative speech reward model for speech RLHF](#). *Preprint*, arXiv:2602.13891.
- Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. 2023. [A long way to go: Investigating length correlations in RLHF](#). *Preprint*, arXiv:2310.03716.
- Hui Wang, Jinghua Zhao, Yifan Yang, Shujie Liu, Junyang Chen, Yanzhe Zhang, Shiwan Zhao, Jinyu Li, Jiaming Zhou, Haoqin Sun, Yan Lu, and Yong Qin. 2026a. [Speechllm-as-judges: Towards general and interpretable speech quality evaluation](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7675–7700, San Diego, California, United States. Association for Computational Linguistics.
- Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. 2023. [Large language models are not fair evaluators](#). *Preprint*, arXiv:2305.17926.
- Siyin Wang, Wenyi Yu, Xianzhao Chen, Xiaohai Tian, Jun Zhang, Lu Lu, Yu Tsao, Junichi Yamagishi, Yuxuan Wang, and Chao Zhang. 2025. [Qualispeech: A speech quality assessment dataset with natural language reasoning and descriptions](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23588–23609, Vienna, Austria. Association for Computational Linguistics.
- Yuanyuan Wang, Dongchao Yang, Yayue Deng, Zhiyong Wu, Steven Y. Guo, Helen M. Meng, and Xixin Wu. 2026b. [UniSRM: A unified speech reward model for reasoning-based fine-grained assessment](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 46346–46366, San Diego, California, United States. Association for Computational Linguistics.

- Mirjam Wester, Cassia Valentini-Botinhao, and Gustav Eje Henter. 2015. [Are we using enough listeners? no! an empirically-supported critique of interspeech 2014 tts evaluations](#). In *Interspeech 2015*, pages 3476–3480. International Speech Communication Association.
- Gijs Wijngaard, Elia Formisano, Michele Esposito, and Michel Dumontier. 2025. [Audsemthinker: Enhancing audio-language models through reasoning over semantics of sound](#). *Preprint*, arXiv:2505.14142.
- Zhifei Xie, Mingbao Lin, Zihang Liu, Pengcheng Wu, Shuicheng Yan, and Chunyan Miao. 2025. [Audio-reasoner: Improving reasoning capability in large audio language models](#). *Preprint*, arXiv:2503.02318.
- Zhen Xiong, Yujun Cai, Zhecheng Li, Junsong Yuan, and Yiwei Wang. 2025. [Thinking with sound: Audio chain-of-thought enables multimodal reasoning in large audio-language models](#). *Preprint*, arXiv:2509.21749.
- Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. 2025. [Qwen2.5-Omni technical report](#). *Preprint*, arXiv:2503.20215.
- Junichi Yamagishi, Christophe Veaux, and Kirsten MacDonald. 2019. [CSTR VCTK Corpus: English multi-speaker corpus for CSTR voice cloning toolkit \(version 0.92\)](#).
- Heiga Zen, Viet Dang, Rob Clark, Yu Zhang, Ron J. Weiss, Ye Jia, Zhifeng Chen, and Yonghui Wu. 2019. [LibriTTS: A corpus derived from LibriSpeech for text-to-speech](#). In *Proceedings of Interspeech*, pages 1526–1530.
- Leying Zhang, Bowen Shi, Haibin Wu, Bach Viet Do, and Yanmin Qian. 2026. [JASTIN: Aligning LLMs for zero-shot audio and speech evaluation via natural language instructions](#). *Preprint*, arXiv:2605.04505.
- Xueyao Zhang, Chaoren Wang, Huan Liao, Ziniu Li, Yuancheng Wang, Li Wang, Dongya Jia, Yuanzhe Chen, Xiulin Li, Zhuo Chen, et al. 2025. [Speech-judge: Towards human-level judgment for speech naturalness](#). *Preprint*, arXiv:2511.07931.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhonghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging LLM-as-a-judge with MT-Bench and chatbot arena](#). *Preprint*, arXiv:2306.05685.
- Kun Zhou, Berrak Sisman, Rui Liu, and Haizhou Li. 2022. [Emotional voice conversion: Theory, databases and ESD](#). *Speech Communication*, 137:1–18.

Appendix

A.1 TTS Systems

Across TTS-generated experiments, we synthesize stimuli with four systems chosen to span autoregressive (AR) and non-autoregressive (non-AR) generation, as well as codec-token and mel-spectrogram output representations (Table 3). The goal is not to compare TTS systems, but to avoid tying the audit to one synthesis architecture.

A.2 Stimulus Pools

Acoustic manipulation pool. The loudness and silence experiments use 80 base utterances filtered from LibriTTS test-clean (Zen et al., 2019). The text filter keeps short declarative sentences, removes questions, exclamations, quotes, abbreviations, loaded tokens, all-caps text, and mid-sentence proper nouns, and preserves a spread over sentence lengths. Each utterance is synthesized once by one of four TTS systems with a VCTK reference speaker (Yamagishi et al., 2019); the 10 reference speakers are gender-balanced (40 female and 40 male assignments; speaker IDs p225, p227, p256, p267, p276, p294, p301, p311, p345, and p363). Intensity adjustments and silence padding are post-hoc waveform transforms on the same base waveform. The silence conditions add 0.5 or 1 second at both the beginning and end of each utterance.

Content-richness pool. The content-richness experiment uses ASSET sentence-simplification pairs (Alva-Manchego et al., 2020). The concise side is a human simplification and the richer side is the original meaning-equivalent Wikipedia sentence. After excluding one catastrophic TTS failure, the analysis uses 149 pairs, approximately balanced

System	Generation structure	Representation
CosyVoice2	AR LLM + flow matching	Codec tokens
Qwen3-TTS	AR LLM + codec decoder	Codec tokens
F5-TTS	Non-AR DiT flow matching	Mel-spectrogram
StyleTTS2	Non-AR diffusion + GAN	Mel-spectrogram

Table 3: TTS systems used for generated-speech stimuli. They are chosen to span autoregressive (AR) and non-autoregressive (non-AR) generation, as well as codec-token and mel-spectrogram output representations.

over TTS systems (38/37/37/37) and speaker gender (75 female, 74 male). Each concise/richer pair is synthesized with the same TTS system, speaker, and reference voice. One representative pair is:

Concise: Perth is used for other settlements worldwide.

Richer: The name Perth has hence been used for a number of other settlement around the world.

Silence-padding and repetition controls are derived from the concise side of the same pair.

Emotion pool. The emotion experiment uses English ESD recordings (Zhou et al., 2022). We use 10 English speakers, 20 texts, and four emotions: Neutral, Happy, Sad, and Angry. All clips are converted to a common sampling format, VAD-trimmed, and normalized to -23 LUFS before scoring. Pairwise emotion trials compare each emotional clip against the matched Neutral clip from the same speaker and text, in both A/B orders.

A.3 Prompts and Reproducibility

Gemini pairwise prompt. For the Gemini pairwise experiments, we use a concatenated-audio setup following AudioJudge-style test-concat evaluation. The system prompt tells the model that the audio contains two speech clips separated by one second of silence and asks it to compare clarity, naturalness, and overall quality. The response is constrained to JSON with a reasoning field and a label:

Gemini Pairwise Speech-Quality Prompt

You are an expert in audio quality assessment specializing in synthesized speech evaluation. The audio you will receive contains TWO speech clips concatenated with 1 second of silence between them: Audio 1 followed by 1 second of silence, then Audio 2. Your task is to critically compare Audio 1 and Audio 2 on clarity, naturalness, and overall quality. ... Respond only with valid JSON with keys `reasoning` and `label`, where `label` is "1", "2", or "tie".

The position/length ablation appends the following line to the system prompt:

Position/Length Prompt Line

Avoid position bias and don't let response length influence your evaluation.

Other judge prompts. For GRM, we use a target-free naturalness-only variant for the content-richness experiment. The released GRM prompt normally includes target text; in the concise/richer setting this would conflate speech quality with text fidelity. For UniSRM, we use its released pointwise prompt for single-clip scoring and a quality-only pairwise prompt for target-free comparisons. UniSRM's Task-AB style prompt includes text-fidelity dimensions; those conditions are excluded from the main content-richness analysis for the same reason as GRM target-text prompts. SQ-LLM is run with its SingleEval and CompareEval formats.

Released audit suite. We release the controlled stimuli, manifests, prompt templates, and analysis scripts used in this study as SPEECHJUDGEAUDIT. The suite allows the same tests to be applied to additional speech judges, including tests of rationale grounding when rationales are available. For ESD, we provide the selected file list and a preparation script rather than redistributing the original recordings.

B.1 Statistical Reporting

For pairwise experiments, the reported effect is the manipulated-clip preference rate, with manipulated-clip wins scored as 1, reference-clip wins as 0, and ties as 0.5. When both A/B and B/A orders are available, we first aggregate the two orders for the same underlying item and then aggregate across items. For pointwise experiments, the reported effect is the within-judge manipulated-reference score difference. Confidence intervals are item-level bootstraps unless otherwise stated. In the full tables, "degenerate" denotes zero variance

Judge	Condition	Pairwise N	Pairwise pref.	Pairwise sig.	Pointwise N	Pointwise Δ	Pointwise sig.	Notes
BTRM	−6 dB vs. ref	—	—	—	100	−0.54	**	quiet penalty
BTRM	−3 dB vs. ref	—	—	—	99	−0.42	**	quiet penalty
BTRM	+3 dB vs. ref	—	—	—	99	+0.05	n.s.	
BTRM	+6 dB vs. ref	—	—	—	99	−0.12	n.s.	
GRM	−6 dB vs. ref	80	0.500	n.s.	—	—	—	pairwise-only judge
GRM	−3 dB vs. ref	80	0.475	n.s.	—	—	—	
GRM	+3 dB vs. ref	80	0.516	n.s.	—	—	—	
GRM	+6 dB vs. ref	80	0.541	*	—	—	—	louder wins
UniSRM	−6 dB vs. ref	80	0.469	n.s.	159	+0.04	n.s.	
UniSRM	−3 dB vs. ref	80	0.481	n.s.	159	+0.01	n.s.	
UniSRM	+3 dB vs. ref	80	0.512	n.s.	159	+0.04	n.s.	
UniSRM	+6 dB vs. ref	80	0.575	**	159	+0.09	**	both methods agree
SQ-LLM	−6 dB vs. ref	80	0.472	n.s.	160	+0.01	n.s.	high pairwise ties
SQ-LLM	−3 dB vs. ref	80	0.484	n.s.	160	+0.01	n.s.	high pairwise ties
SQ-LLM	+3 dB vs. ref	80	0.506	n.s.	160	0.00	deg.	integer-scale collapse
SQ-LLM	+6 dB vs. ref	80	0.591	***	160	0.00	deg.	pairwise reveals preference
Gemini-2.5	−6 dB vs. ref	80	0.431	***	156	−0.09	n.s.	ref wins pairwise
Gemini-2.5	−3 dB vs. ref	80	0.497	n.s.	155	−0.11	n.s.	
Gemini-2.5	+3 dB vs. ref	80	0.503	n.s.	155	−0.23	n.s.	
Gemini-2.5	+6 dB vs. ref	80	0.566	***	155	−0.13	n.s.	pairwise reveals direction
Gemini-3.1	−6 dB vs. ref	80	0.384	***	155	−0.15	n.s.	monotonic pairwise pattern
Gemini-3.1	−3 dB vs. ref	80	0.481	n.s.	155	−0.16	n.s.	
Gemini-3.1	+3 dB vs. ref	80	0.528	*	155	−0.09	n.s.	louder wins pairwise
Gemini-3.1	+6 dB vs. ref	80	0.616	***	154	−0.19	n.s.	strongest pairwise effect

Table 4: Combined loudness results. Pairwise preference is $P(\text{manipulated} > \text{reference})$ with ties counted as 0.5; pointwise Δ is the within-judge manipulated–reference score difference. All pairwise judges prefer +6 dB clips, while pointwise scoring often hides this preference.

Judge	Condition	Pairwise N	Pairwise pref.	Pairwise sig.	Pointwise N	Pointwise Δ	Pointwise sig.	Notes
BTRM	pad500 vs. base	—	—	—	80	−1.59	*	pad penalty
BTRM	pad1000 vs. base	—	—	—	80	−2.87	***	pad penalty grows
UniSRM	pad500 vs. base	80	0.250	***	80	−0.01	n.s.	pointwise 84% identical
UniSRM	pad1000 vs. base	80	0.069	***	80	+0.01	n.s.	pointwise 86% identical
SQ-LLM	pad500 vs. base	80	0.537	n.s.	80	+0.01	n.s.	96% identical scores
SQ-LLM	pad1000 vs. base	80	0.422	n.s.	80	−0.04	n.s.	trend anti-pad
Gemini-2.5	pad500 vs. base	80	0.512	n.s.	80	+0.59	**	pointwise pad preference
Gemini-2.5	pad1000 vs. base	80	0.506	n.s.	80	+0.49	*	pairwise low-power
Gemini-3.1	pad500 vs. base	80	0.503	n.s.	80	−0.24	n.s.	pairwise low-power
Gemini-3.1	pad1000 vs. base	80	0.509	n.s.	80	−0.32	n.s.	pairwise low-power

Table 5: Combined silence-padding results. Pointwise Δ is the padded–base score difference; pairwise pad preference is $P(\text{padded} > \text{base})$ with ties counted as 0.5. UniSRM illustrates the diagnostic pattern most clearly: pointwise scores are nearly unchanged, but pairwise comparison strongly rejects padding.

or integer-scale collapse, and “low-power” marks high-tie pairwise settings with too few decisive outcomes. Significance markers are * ($p < .05$), ** ($p < .01$), and *** ($p < .001$); “n.s.” denotes a non-significant result.

B.2 Loudness and Silence Results

Table 4 reports loudness effects side by side: every pairwise judge prefers +6 dB, while pointwise often hides it. Table 5 shows the analogous silence result: UniSRM pointwise is nearly unchanged, but pairwise rejects padding.

B.3 Content-Richness Results

Table 6 shows that an explicit position/length instruction does not remove Gemini-3.1’s content-

Setting	Richer preference	Takeaway
Baseline	0.701	reference
Position/length instruction	0.695	preference remains

Table 6: Gemini-3.1 position/length prompt ablation on ASSET content-richness pairs. The explicit instruction has negligible effect: the richer-speech preference remains near 0.70.

richness preference. Table 7 shows that richer speech is preferred by every pairwise judge, while silence-padding and repetition do not reproduce the effect. Text-fidelity controls are needed because target transcripts can confound speech quality with transcript matching (Table 8).

Judge	Richer > concise	Sig.	Pad > concise	Sig.	Repeat > concise	Sig.	Richer > pad	Sig.
UniSRM	0.846	***	0.082	***	0.447	**	0.954	***
GRM	0.570	***	0.527	n.s.	0.362	***	0.567	***
SQ-LLM	0.755	***	0.424	*	0.515	n.s.	0.765	***
Gemini-2.5	0.664	***	0.507	n.s.	0.515	n.s.	0.544	**
Gemini-3.1	0.701	***	0.502	n.s.	0.471	***	0.661	***

Table 7: Full content-richness decomposition. Values are manipulated-clip preference rates with ties counted as 0.5. Richer denotes the original ASSET sentence, C = concise, Pad = silence-padded concise, and Rep = repeated concise; significance markers follow Appendix B.1.

Judge	Setting	Text fidelity?	Richer pref.	What it shows
BTRM	Concise target transcript supplied	yes	0.705	target transcript changes the outcome
BTRM	Richer target transcript supplied	yes	0.262	target mismatch reverses the direction
BTRM	Concise and richer targets averaged	yes	0.483	target effects cancel; no target-free analogue
UniSRM	Target transcript plus text-fidelity dimension	yes	–	text fidelity masks content-richness preference
UniSRM	Target-free prompt variant	no	–	output became template-like, so excluded
Gemini-3.1	Anti-position/length instruction only	no	0.695	preference remains despite the instruction

Table 8: Text-fidelity and prompt controls for the content-richness experiment. Target-text conditions can confound acoustic preference by rewarding or penalizing extra words through transcript matching.

B.4 Emotional Delivery Results

The emotion analysis first asks whether pointwise scores respond to emotional delivery at all. For each judge, we z-score scores within judge and fit a mixed-effects model with emotion as a fixed effect and random intercepts for speaker and text:

$$\text{score_z} \sim \text{emotion} + (1 \mid \text{speaker}) + (1 \mid \text{text}).$$

Here `score_z` is the judge score z-scored within judge; `emotion` uses Neutral as the reference level; and the two random intercepts account for repeated speakers and repeated texts. The pointwise β values in Table 9 are the estimated emotion contrasts against Neutral. The overall significance column comes from a likelihood-ratio test comparing this model against one without the emotion fixed effect. Four of the five pointwise judges show a significant overall emotion effect, indicating that their scores are sensitive to emotional delivery.

The second question is whether this sensitivity becomes a directional quality preference under pairwise comparison. Table 9 compares each emotional clip against the matched Neutral clip from the same speaker and text. Sad loses to Neutral for four of five pairwise judges, with UniSRM as the exception. Happy is above chance for UniSRM, SQ-LLM, and Gemini-3.1, but flat for GRM and Gemini-2.5. Thus judges do not merely perceive emotional variation; several map it into quality preferences.

B.5 Pointwise and Pairwise Comparison

Table 10 summarizes where pairwise comparison reveals directional preferences that pointwise scores obscure.

B.6 Position-Bias Diagnostics

Pairwise trials are order-counterbalanced because several judges show strong presentation-order effects. These diagnostics are not treated as acoustic shortcuts, but they are important for interpreting pairwise preferences: without A/B and B/A averaging, a position effect could be mistaken for an acoustic preference.

C.1 Rationale Audits

The rationale audit separates behavior from explanation. The key pattern is that rationales often use generic rubric language while failing to name the manipulated acoustic cue. A representative pairwise example is:

Judge	Emotion	Pairwise N	Pairwise pref.	Pairwise sig.	Pointwise β	Pointwise sig.	Overall sig.	Notes
BTRM	Happy	—	—	—	-0.085	n.s.	**	
BTRM	Sad	—	—	—	-0.205	***	**	lower than Neutral
BTRM	Angry	—	—	—	-0.129	*	**	lower than Neutral
GRM	Happy	400	0.478	n.s.	—	—	—	flat
GRM	Sad	400	0.392	***	—	—	—	Neutral wins
GRM	Angry	400	0.395	***	—	—	—	Neutral wins
UniSRM	Happy	400	0.598	***	-0.040	n.s.	***	Happy wins pairwise
UniSRM	Sad	400	0.555	*	+0.072	n.s.	***	pro-Sad outlier
UniSRM	Angry	400	0.535	n.s.	-0.392	***	***	pointwise penalty
SQ-LLM	Happy	400	0.551	*	-0.126	n.s.	***	Happy wins pairwise
SQ-LLM	Sad	400	0.328	***	-0.427	***	***	strongest Sad penalty
SQ-LLM	Angry	400	0.473	n.s.	-0.142	n.s.	***	flat pairwise
Gemini-2.5	Happy	400	0.470	n.s.	-0.267	**	n.s.	mixed evidence
Gemini-2.5	Sad	400	0.366	***	-0.248	**	n.s.	Neutral wins
Gemini-2.5	Angry	400	0.429	**	-0.310	***	n.s.	Neutral wins
Gemini-3.1	Happy	400	0.558	*	+0.250	**	***	Happy wins
Gemini-3.1	Sad	400	0.372	***	-0.272	**	***	Neutral wins
Gemini-3.1	Angry	400	0.482	n.s.	+0.188	*	***	pointwise positive

Table 9: Combined emotion results. Pairwise preference is $P(\text{emotion} > \text{Neutral})$ with ties counted as 0.5; pointwise β is the z-scored coefficient relative to Neutral. Overall significance is from the likelihood-ratio test for the emotion fixed effects.

Cue / example	Pointwise evidence	Pairwise evidence	Takeaway
Content richness	no consistent richer-speech advantage; Gemini-2.5 leans concise	richer preference 0.570–0.846	pairwise comparison reveals a consistent preference
Content richness correlations	Gemini-3.1 $r = +0.14$; SQ-LLM $r = +0.096$	same items, pairwise preferences	per-item pointwise and pairwise verdicts are weakly related
Silence / UniSRM	$\Delta \approx 0$, 84–86% identical scores	pad1000 preference 0.069	integer scale hides anti-pad preference
Loudness / SQ-LLM	loud6 $\Delta = 0.000$, degenerate	loud6 preference 0.591	pointwise resolution ceiling hides pro-loud preference
Loudness / Gemini	all pointwise deltas n.s.	quiet6 loses, loud6 wins	pointwise noise hides directional level preference
Emotional delivery	Kendall $W = 0.33$ across pointwise ranks	Kendall $W = 0.84$ across pairwise ranks	pairwise rankings are more consistent across judges

Table 10: Pointwise and pairwise evidence across controlled acoustic manipulations. Pairwise comparison often reveals a directional preference that is weak, noisy, or absent in pointwise scores.

Rationale example: Sad vs. Neutral

GRM pairwise output, excerpted verbatim. ESD speaker 0012 reads “One two three, where is your breakfast” in Neutral (A) and Sad (B) delivery.

Output A: “The intonation is very natural and human-like... The pacing is smooth and consistent, mirroring natural speech... This output sounds remarkably natural, almost indistinguishable from a human speaker.”

Output B: “The prosody is quite unnatural... very flat and robotic... The pacing is slow and deliberate, with noticeable and unnatural pauses... This output sounds highly artificial and robotic.”

Conclusion: Output A: 9; Output B: 2.

The rationale identifies several audible differences, but treats them as naturalness defects without contextualizing them as Sad delivery before assigning the lower score.

Figure 5 summarizes the same pattern across manipulations. Terms naming the manipulated acoustic change are rare for loudness, silence, duration, and emotion. Repetition is the partial exception: some judges can name it, but detection remains incomplete and does not fully determine the penalty.

The explicit-cue lexicons are narrow by design. For intensity, we count direct references to loudness, volume, quietness, amplitude, level, or dB. For boundary silence, we count silence, padding, leading/trailing silence, pause, and gap terms when they refer to the clip boundary. For content richness, we count length, verbosity, richer content, extra words, and related wording. For emotional delivery, we count emotion-specific words for Happy, Sad, Angry, and Neutral, while excluding generic prosody terms such as “tone,” “intonation,” “prosody,” and “expressive.” For repetition, we count repeat, repetition, duplicate, and related terms.

D.1 Human Preference Calibration

The human preference calibration uses 15 listeners with native or near-native English proficiency. Each listener was assigned 30 pairs and responded with A, B, or tie, yielding 449 valid judgments. The 60 unique pairs comprise 12 loudness pairs,

Setting	Position effect observed	Role in this audit
Content richness	GRM strongly favors the second clip in decisive comparisons (B wins 91.6%); Gemini-2.5 also favors B (71.8% of trials), while UniSRM and SQ-LLM lean toward A (64.8% and 56.4%).	We average the two presentation orders before computing the richer-speech preference, so the reported effect is not a single-order artifact.
Loudness	GRM shows order asymmetries of roughly 10–15 percentage points across loudness conditions even when its order-averaged loudness preference is near chance.	Order averaging prevents a positional preference from being interpreted as evidence for or against a loudness preference.
Silence padding	GRM and SQ-LLM show clear AB/BA differences, while their order-averaged padded-clip preferences remain near chance.	This supports using order-averaged item preferences and treating the strongest silence result as judge-specific rather than universal.
Emotional delivery	The average AB–BA gap is about 0.10 for SQ-LLM, 0.48 for GRM, 0.61 for Gemini-2.5, 0.68 for Gemini-3.1, and 0.80 for UniSRM.	The emotion preferences in Table 9 are interpretable because they are averaged across both orders.

Table 11: Position-bias diagnostics for pairwise experiments. The numbers are used to justify order counterbalancing, not as acoustic-shortcut results.

Manipulation	Judge(s) audited	Explicit acoustic-term rate	Generic rubric terms
Content richness / length	Gemini-3.1, UniSRM	GRM, length 0–3%; richer/original 0%	rubric/prosody terms frequent
Loudness	UniSRM	loud/loudness/volume 0%; amplitude/dB/level 0.94%	prosody/naturalness/quality 100%
Silence padding	UniSRM	silence 0%; duration 0%; pad/leading/trailing 1.25%	prosody/naturalness/quality 100%
Emotion	rationale-producing judges	emotion-specific terms rarely identify the true emotion (3/16 pointwise cells)	generic prosody terms excluded from lexicon
Repetition	GRM	repetition keywords 22.1% for rep-vs-concise	partial exception; detection incomplete

Table 12: Summary of explicit acoustic-term rates in rationales. Generic rubric terms refer to broad quality or prosody language rather than the manipulated acoustic change.

18 content-richness pairs, 15 Sad-vs-Neutral pairs, and 15 Happy-vs-Neutral pairs. The instruction is: “These clips are synthesized or recorded speech samples. For each pair, choose which utterance has better overall naturalness and speech quality.” Table 14 combines manipulated-clip preference rates, human significance tests, judge residuals, and agreement diagnostics. Entropy is computed over the item-level distribution of {manipulated, tie, reference}; higher entropy and variance indicate more listener disagreement. Human brackets are bootstrap confidence intervals over items. Judge brackets are ranges over the five pairwise judges, so they describe model spread rather than sampling uncertainty. Table 15 reports listener-level quality-control diagnostics.

Table 14 supports three interpretations used in the main text. Higher loudness is perceptually salient for humans, but remains invalid evidence of better TTS quality for a judge. Content richness is near chance for humans but consistently preferred by judges. For emotion, Sad-vs-Neutral has strong

listener agreement in favor of Neutral, whereas Happy-vs-Neutral is more split for humans while three of five judges significantly favor Happy.

E.1 Supporting Analyses

Table 16 reports supporting diagnostics for the scope of the audit. They show which attributes did not reliably change judgments within our pools, and which associations remain diagnostic rather than causal.

Judge	Mentions rep.	$P(\text{rep loses} \mid \text{mentioned})$	$P(\text{rep loses} \mid \text{not})$
Gemini-2.5	5.7%	11.8%	13.9%
Gemini-3.1	27.9%	16.9%	2.3%
GRM	21.8%	78.5%	59.7%
UniSRM	10.5%	80.6%	45.1%

Table 13: Repetition detection versus penalty. Mentions rep. is the fraction of rep-vs-concise rationales mentioning repetition keywords; the conditional columns report how often the repeated clip loses when repetition is or is not named.

Condition	Items	N	Human mean [CI]	Human sig.	Judge mean [range]	Sig. judges	Residual	Human entropy	Human variance
+6 dB vs. ref	12	88	0.612 [0.548, 0.671]	*	0.578 [0.541, 0.616]	5/5	-0.035	1.009	0.071
Richer vs. concise	18	134	0.547 [0.464, 0.627]	n.s.	0.707 [0.570, 0.846]	5/5	+0.160	1.260	0.111
Sad vs. Neutral	15	115	0.217 [0.129, 0.320]	***	0.403 [0.328, 0.555]	5/5	+0.186	0.756	0.098
Happy vs. Neutral	15	112	0.407 [0.326, 0.488]	n.s.	0.531 [0.470, 0.598]	3/5	+0.124	1.305	0.155

Table 14: Human preference calibration and agreement diagnostics. Human mean is the item-level manipulated-clip preference rate with bootstrap 95% confidence intervals. Human significance uses a two-sided sign-flip test against 0.5. Judge mean is averaged over the five pairwise judges, the bracketed range shows their spread, and Sig. judges counts how many individual judges are significant in either direction.

Rater	Valid N	Mean pref.	Tie rate	Majority agreement	Item-mean MAE
Lowest-agreement rater (rater_14)	29	0.379	0.207	0.345	0.478
Most extreme rater (rater_11)	30	0.450	0.100	0.433	0.465
All raters mean	29.9	0.441	0.343	0.481	0.311

Table 15: Human listener quality-control diagnostics. Majority agreement is the fraction of a rater’s responses that match the item-level majority label; item-mean MAE is the mean absolute error between the rater’s numeric response and the item mean after coding manipulated/reference/tie as 1/0/0.5.

Analysis	Statistic	Result
Speaker gender	female vs. male speakers	no preference (n.s.; all $p \geq .21$)
Speaker accent	North-American vs. British speakers	no preference (n.s.)
Emotion gender interaction	gender $\times$ emotion	no interaction (n.s.)
Low-level descriptors	descriptor–score correlations	no corrected correlation (n.s.)
Model self-preference	Qwen judges on Qwen3-TTS vs. other TTS	no self-preference (n.s.)
Gemini TTS preference	TTS-system effect	preference differs by system (***)
Gemini-2.5 duration	duration–score correlation	negative correlation (***)
SQ-LLM rubric coupling	Emotional Impact vs. Overall Quality	low Emotional Impact is associated with more Overall=3 ratings (+16 pp; diagnostic)
Gemini rationale wording	expressive vs. monotone wording	score gap $\approx 2.5\text{--}2.75/10$ (diagnostic)

Table 16: Natural-variance and supporting analyses. These diagnostics are within-pool evidence and should not be read as causal mechanisms or broad fairness guarantees.